

MUHAMMAD SAQLAIN

Researcher — ML, CV, Open Source Systems, Educator
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ACHIEVEMENTS

2× Google Summer of Code (top ~5%): Selected twice for GSoC — 2026, NumFOCUS/DeepForest (4.88%; recovering historical forest data for Oregon with University of Florida) and 2025, Google Chromium (5.43%; earned official saqlain@chromium.org status).

Open-Source Mentorship: Mentored underprivileged Pakistani students for GSoC/ESoC/LFx/OSPP via Dev Weekends; 11 selected in 2026 (9 GSoC, 1 ESoC, 1 LFx); mentees earned \$50,138+ in stipends over 2 years.

Research with a Top-Ranked Scientist: Working with the #1-ranked Computer Science researcher from Pakistan (now at PMU, Saudi Arabia) on deep-learning CNNs for remote sensing.

Stanford Code in Place (2×): Section Leader for Stanford's Code in Place (~16,000 students); taught CS106A to 26 students from 16 countries over 2 years. Credential

Competitive Programming: ICPC Asia Topi Regional 2024; Bronze at CALICO 2024 (UC Berkeley) (252/850+ teams); 73rd/500+ at MIT Informatics Tournament 2025; 200+ problems on LeetCode/Codeforces.

EDUCATION

University of Education <i>MS in Information Technology</i> CGPA: 3.85/4.00	Lahore, Pakistan <i>August 2025 – 2027</i>
Coursework: Research Methodology, Advanced Information Security, Advanced Topics in Computing, Advanced Computer Networks, Mobile and Wireless Networks, Internet of Things (IoT)	
Standardized Tests: IELTS — Overall 7.5 (C1 Advanced) <u>score report</u>	

University of Education <i>Bachelors in Information Technology</i> CGPA: 3.13/4.00	Lahore, Pakistan <i>Dec 2021 – May 2025</i>
Coursework: OOP, DBMS, Algebra, OS, DSA, System and Network Administration, AI, Computer Networks, Discrete Math, Statistics and Probability	

PUBLICATIONS & MANUSCRIPTS IN PREPARATION

Saqlain, M., Veitch-Michaelis, J., Senyondo, H., Weinstein, B., & White, E. *Automated Ortho-Registration of Historical U.S. Forest Service Aerial Detection Survey Data to NAIP Imagery for Forest-Health Monitoring*. (Manuscript in preparation — Weecology Lab, University of Florida.)

Saqlain, M., Abbas, M. J., & Khan, M. A. *Deep Learning for Remote-Sensing-Based Crop-Disease Diagnosis and Precision Agriculture*. (Manuscript in preparation — Center for Artificial Intelligence, Prince Mohammad Bin Fahd University.)

RESEARCH EXPERIENCE

GSoC '26 Research Contributor <i>Weecology Lab, University of Florida – NumFOCUS / DeepForest</i>	Gainesville, FL (Remote) <i>May 2026 – Present</i>
- Mentored by Dr. Ethan White (PI), Ben Weinstein & Josh Veitch-Michaelis (Research Scientists), and Henry Senyondo (Research Software Engineer) on <u>Recovering Historical Image Data Using Automated OrthoRegistration</u> (4.88% selection rate). - Building an automated pipeline to align U.S. Forest Service Aerial Detection Survey (ADS) polygons with year-matched NAIP satellite imagery, correcting 50–500m spatial offsets caused by GPS imprecision and flight geometry errors. - Developed a hybrid alignment algorithm (ProximityAlign adaptation) combining inside-outside contrast energy, contour-to-boundary distance, and out-of-bounds penalty; improved mean IoU from 0.082 to 0.231 on seed pairs. - Scaling the pipeline to process ~48,000 Oregon ADS polygons and fine-tuning DeepForest for NAIP-based forest health detection. - Conducted a <u>comparative benchmarking study</u> of DeepForest vs DeepTrees, analyzing architectural differences (RetinaNet vs U-Net+watershed), prediction outputs (bounding boxes vs polygons), vegetation indices, and fine-tuning workflows; presented findings to mentors during the selection phase.	

Research Associate <i>Center for Artificial Intelligence (CAI), Prince Muhammad Bin Fahd University</i>	Al Khobar, Saudi Arabia <i>Jan. 2026 – Present</i>
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- Building computer-vision pipelines for **remote sensing** and **precision agriculture**: multi-class classification, semantic segmentation, and object detection on satellite and field imagery.
- Developing deep-learning models for crop-disease diagnosis and satellite-image enhancement, benchmarking **Vision Transformer (ViT)** backbones against hybrid CNN–Transformer architectures.
- Running the data, training, and evaluation pipeline toward a first-author manuscript on remote-sensing-based crop-disease detection.

GSoc'25 Chromium OS Contributor

Google Summer of Code (GSoc) – Web Audio Team

Remote

May 2025 – Nov. 2025

- Selected for the Google Summer of Code program (5.43% selection rate) to modernize the testing infrastructure of the Chromium browser engine.
- Engineered a Generative AI automation prototype (Chromium-Automate-CLs) using Gemini models; the tool automated ~90% of test conversion tasks and was highlighted by mentors as a roadmap for future AI-assisted workflows in Chromium.
- Designed a Python heuristic analysis system to parse 326 complex test files based on "Code Complexity Scores" (AST parsing of loops, async logic), creating the data-driven strategy for the migration.
- Delivered 96 CLs (~95% merge rate) to migrate the Web Audio test suite to the W3C `testharness.js` framework, enforcing Google's strict JS style rules and optimizing CI bandwidth usage by ~187 GB/day.

TEACHING EXPERIENCE (volunteer)

Section Leader (2× Selection)

Stanford University: Code in Place

California, United States (Remote)

Apr 2025 – Jun 2025, Apr 2026 – Present

- Selected twice for Stanford's Code in Place, one of the most diverse CS courses globally (~16,000 enrolled students).
- **2026:** Selected from ~2,000 candidates to teach CS106A Python to 11 students from 8 countries: UK (3), Venezuela (2), Serbia, Germany, Peru, Czech Republic, Russia, USA.
- **2025:** Taught Python to ~15 international students from 11 countries: USA (5), Somalia, Thailand, Chile, Philippines, Germany, Ecuador, India, Peru, Canada. Promoted to Section Leader Mentor for 1-on-1 support and taught 6 TeachNow sessions. Credential
- Over 2 years, taught 26 students from 16 countries.

Mentor

Dev Weekends

Remote

May 2024 – Present

- Initiated and led a 6-month open-source mentoring drive, running weekly sessions, proposal reviews, and 1-on-1 guidance for 275+ students on competitive programs (GSoc, ESoC, LFX, OSPP).
- **2026:** 11 students selected (9 GSoc, 1 ESoC, 1 LFX). **2025:** 3 students selected (2 GSoc, 1 OSPP). Collectively, mentees earned over \$50,000 in stipends.
- Delivered 25+ lectures on LeetCode and core DSA, teaching arrays, linked lists, trees, graphs, sorting, searching, and dynamic programming.

Trainer and Moderator

iCodeGuru

Santa Clara, California Remote

Feb 2023 – Present

- Offered volunteer teaching in Python and DSA at iCodeGuru and solved numerous LeetCode problems. All recording are here
- Taught courses on the IELTS speaking and writing modules specifically.

PROJECTS

Your JobBuddy (Final Year Project) GitHub — Sept 2024 – May 2025

- Developed a **GenAI-powered resume optimization platform** using Next.js (TypeScript) to maximize ATS (Applicant Tracking System) compatibility for job seekers.
- Integrated LLMs to implement resume parsing, automated scoring, and context-aware feedback on clarity and formatting.
- Designed a responsive interface featuring real-time document editing, multi-template support, and PDF generation.

QuillPDF Live Demo

- Engineered a **RAG-based (Retrieval-Augmented Generation)** SaaS using LangChain, Pinecone, and OpenAI to enable AI conversations with PDF documents.

- Built a high-performance full-stack architecture with Next.js, Prisma, and Tailwind CSS, supporting free/pro subscription tiers.

ClearOUT (AI Hackathon) [Live](#) — [Submission](#)

- Led a 6-member international team in the lablab.ai LangFlow Hackathon to build an AI-powered recycling/reuse assistant using image recognition and LangChain/LangFlow.

TECHNICAL SKILLS

Programming Languages: Python, JavaScript, TypeScript, C++

ML & Deep Learning: PyTorch, scikit-learn, NumPy, SciPy, Pandas, OpenCV

Computer Vision & Remote Sensing: Object Detection, Image Registration, GeoPandas, Rasterio, GDAL, QGIS

Web Dev: Next.js, Node JS, Express JS

DevOps & Tools: Docker, CI/CD, GitHub Actions, AWS (RDS, S3), Git

AI: Generative AI, LLMs, RAG (LangChain, Pinecone)

Databases: MongoDB, MySQL, PostgreSQL

REFERENCES

Dr. Ethan White — Professor & PI, Weecology Lab @ University of Florida — [LinkedIn](#)

- Email: ethanwhite@ufl.edu
- Role: GSoC 2026 Mentor (NumFOCUS / DeepForest)
- Location: Gainesville, FL, United States

Dr. Ben Weinstein — Research Scientist, Weecology Lab @ University of Florida — [LinkedIn](#)

- Email: ben.weinstein@weecology.org
- Role: GSoC 2026 Mentor (NumFOCUS / DeepForest)
- Location: Gainesville, FL, United States

Dr. Josh Veitch-Michaelis — Research Scientist (Postdoc), Weecology Lab @ University of Florida — [LinkedIn](#)

- Email: j.veitchmichaelis@gmail.com
- Role: GSoC 2026 Mentor (NumFOCUS / DeepForest)
- Location: Gainesville, FL, United States

Henry Senyondo — Research Software Engineer, Weecology Lab @ University of Florida — [LinkedIn](#)

- Email: henrykironde@gmail.com
- Role: GSoC 2026 Mentor (NumFOCUS / DeepForest)
- Location: Gainesville, FL, United States

Hongchan Choi — Tech Lead & Manager, Google (Chromium) — [LinkedIn](#)

- Email: hongchan@google.com
- Web Audio / Web MIDI API Owner: Chromium Project
- Adjunct Professor: Stanford University
- Location: Mountain View, CA

Dr. Muhammad Attique Khan — Assistant Research Professor, Center of AI @ PMU — [LinkedIn](#)

- Email: mkhan3@pmu.edu.sa
- Location: Saudi Arabia